

Graduate-Level Notes on Maxwell's Equations

Differential and Integral Forms, Derivations, Consistency Proofs, Potentials, Waves,
Boundary Conditions, and Worked Examples

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1 Introduction

Maxwell's equations are the fundamental equations of classical electromagnetism. They unify electrostatics, magnetostatics, electromagnetic induction, charge conservation, and light. In SI units, the vacuum Maxwell equations are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (4)$$

These equations must be supplemented by the Lorentz force law,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (5)$$

which tells us how fields act on matter.

Remark 1.1. *A very important conceptual point is the following:*

- *Maxwell's equations are physical laws, not theorems derived from pure mathematics.*
- *What can be proved mathematically, once the equations are assumed, are:*
 1. *the equivalence of their integral and differential forms,*
 2. *charge conservation,*
 3. *the existence of electromagnetic waves,*
 4. *boundary conditions,*
 5. *the existence of potentials,*
 6. *energy and momentum conservation laws.*

These notes are written in a graduate style: we will derive the equations from physical input where possible, prove equivalences carefully, and derive the central mathematical and physical consequences.

2 Notation and Mathematical Preliminaries

We use:

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &:= \text{electric field,} \\ \mathbf{B}(\mathbf{r}, t) &:= \text{magnetic field,} \\ \rho(\mathbf{r}, t) &:= \text{charge density,} \\ \mathbf{J}(\mathbf{r}, t) &:= \text{current density,} \\ \phi(\mathbf{r}, t) &:= \text{scalar potential,} \\ \mathbf{A}(\mathbf{r}, t) &:= \text{vector potential.} \end{aligned}$$

The vacuum constants satisfy

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}. \quad (6)$$

2.1 Integral Theorems

We will repeatedly use the following.

Divergence theorem

For a smooth vector field $\mathbf{F}$ on a volume V with boundary ∂V ,

$$\iiint_V \nabla \cdot \mathbf{F} dV = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{a}. \quad (7)$$

Stokes' theorem

For a smooth vector field $\mathbf{F}$ on an oriented surface S with boundary ∂S ,

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = \oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell}. \quad (8)$$

2.2 Vector Identities

We will need

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0, \quad (9)$$

$$\nabla \times (\nabla f) = 0, \quad (10)$$

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}, \quad (11)$$

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}). \quad (12)$$

3 The Four Equations: Names, Differential and Integral Forms

Maxwell's equations in vacuum

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad \Longleftrightarrow \quad \oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}, \quad (13)$$

$$\nabla \cdot \mathbf{B} = 0 \quad \Longleftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{a} = 0, \quad (14)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \Longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}, \quad (15)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad \Longleftrightarrow \quad \oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a}. \quad (16)$$

The names are:

1. **Gauss's law for electricity,**
2. **Gauss's law for magnetism,**
3. **Faraday's law of induction,**
4. **Ampère-Maxwell law.**

The first two are divergence equations, so they describe *sources* of fields. The last two are curl equations, so they describe *circulation* and dynamical coupling.

4 Gauss's Law for Electricity

4.1 Physical Statement

Electric charge is the source of electric field. Positive charges produce outward electric flux; negative charges produce inward flux.

4.2 From Coulomb's Law to Gauss's Law

For a point charge q at the origin, Coulomb's law gives

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}. \quad (17)$$

Take a sphere of radius r centered on the charge. Then $\mathbf{E}$ is everywhere normal to the surface and has constant magnitude

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}. \quad (18)$$

Therefore,

$$_S \mathbf{E} \cdot d\mathbf{a} = E_S da \quad (19)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} (4\pi r^2) \quad (20)$$

$$= \frac{q}{\epsilon_0}. \quad (21)$$

So for a point charge,

$$_S \mathbf{E} \cdot d\mathbf{a} = \frac{q}{\epsilon_0}. \quad (22)$$

By superposition, for any charge distribution,

$$_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (23)$$

which is the integral form of Gauss's law.

4.3 Differential Form

Applying the divergence theorem,

$$\iiint_V \nabla \cdot \mathbf{E} dV = \oint_{\partial V} \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \iiint_V \rho dV. \quad (24)$$

Hence

$$\iiint_V \left(\nabla \cdot \mathbf{E} - \frac{\rho}{\epsilon_0} \right) dV = 0 \quad (25)$$

for every volume V , so

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}} \quad (26)$$

4.4 Interpretation

The divergence $\nabla \cdot \mathbf{E}$ measures the local source strength of the electric field. This is the precise local version of “charge generates electric field.”

5 Gauss's Law for Magnetism

5.1 Physical Statement

Experimentally, magnetic field lines do not begin or end on isolated magnetic charges. Instead, magnetic field lines form closed loops or extend continuously through space. Thus the net magnetic flux through any closed surface is zero:

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0. \quad (27)$$

5.2 Differential Form

By the divergence theorem,

$$\iiint_V \nabla \cdot \mathbf{B} dV = \oint_{\partial V} \mathbf{B} \cdot d\mathbf{a} = 0. \quad (28)$$

Since this holds for every volume,

$$\boxed{\nabla \cdot \mathbf{B} = 0.} \quad (29)$$

5.3 Interpretation

A divergence-free field has no sources or sinks. That is why $\mathbf{B}$ may be represented in terms of a vector potential:

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (30)$$

This representation is central to advanced electrodynamics, quantum mechanics, and gauge theory.

6 Faraday's Law of Induction

6.1 Physical Statement

A changing magnetic field induces an electric field. In integral form, for a fixed loop C bounding a surface S ,

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}. \quad (31)$$

The minus sign is Lenz's law: the induced circulation opposes the change in magnetic flux.

6.2 Differential Form

Using Stokes' theorem,

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a}. \quad (32)$$

Hence,

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}. \quad (33)$$

For a fixed surface,

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = \iint_S \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{a}. \quad (34)$$

Therefore,

$$\boxed{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.} \quad (35)$$

6.3 Interpretation

This equation says that electric fields are not always conservative. In electrostatics, $\nabla \times \mathbf{E} = 0$, so $\mathbf{E} = -\nabla\phi$. But in time-dependent situations, $\nabla \times \mathbf{E} \neq 0$, so there need not exist a globally single-valued electrostatic potential description alone.

7 Ampère's Law and Maxwell's Correction

7.1 Magnetostatic Ampère Law

For steady currents,

$$\oint_C \mathbf{B} \cdot d\ell = \mu_0 I_{\text{enc}}, \quad (36)$$

or locally,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (37)$$

This is correct in magnetostatics.

7.2 Why It Fails Dynamically

Take the divergence:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \nabla \cdot \mathbf{J}. \quad (38)$$

But by (9),

$$0 = \mu_0 \nabla \cdot \mathbf{J}. \quad (39)$$

So the original equation would force

$$\nabla \cdot \mathbf{J} = 0. \quad (40)$$

That is too restrictive. In general, local charge conservation requires

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (41)$$

Thus if charge density changes in time, the simple magnetostatic equation fails.

7.3 Maxwell's Idea

From Gauss's law,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}. \quad (42)$$

Differentiate with respect to time:

$$\nabla \cdot \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\varepsilon_0} \frac{\partial \rho}{\partial t}. \quad (43)$$

Using the continuity equation,

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}, \quad (44)$$

so

$$\nabla \cdot \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = 0. \quad (45)$$

Therefore the consistent generalization is

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}}. \quad (46)$$

7.4 Integral Form

Integrating over a surface and applying Stokes' theorem gives

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a}. \quad (47)$$

The term

$$I_D := \varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a} \quad (48)$$

is the **displacement current**.

7.5 Capacitor Argument

This is most vividly seen in a charging capacitor. A wire carries current into one plate and out of the other. If one chooses a surface cutting through the wire, one finds current I_{enc} . If one chooses instead a bulging surface spanning the same loop but passing between the plates, there is no conduction current through that surface. Without the displacement current term, Ampère's law would give two different answers for the same loop. Maxwell's correction fixes this and makes the law surface-independent.

8 Proof of Equivalence Between Integral and Differential Forms

Proposition 8.1. *Assume the fields are sufficiently smooth. Then the differential and integral forms of Maxwell's equations are equivalent.*

Proof. We prove two representative cases; the others are identical in structure.

Case 1: Gauss's law for electricity.

Assume the integral form:

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\varepsilon_0} \iiint_V \rho dV \quad (49)$$

for all volumes V . By the divergence theorem,

$$\iiint_V \nabla \cdot \mathbf{E} dV = \frac{1}{\varepsilon_0} \iiint_V \rho dV. \quad (50)$$

Hence

$$\iiint_V (\nabla \cdot \mathbf{E} - \rho/\varepsilon_0) dV = 0 \quad (51)$$

for every volume V , implying

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}. \quad (52)$$

Conversely, integrating the differential form over any volume and applying the divergence theorem gives the integral form.

Case 2: Faraday's law.

Assume the integral form:

$$\oint_{\partial S} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}. \quad (53)$$

By Stokes' theorem,

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = \iint_S \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{a}. \quad (54)$$

Since this holds for arbitrary surfaces,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (55)$$

Conversely, integrate the differential form over a surface and use Stokes' theorem to recover the integral form. $\square$

9 Charge Conservation

Charge conservation is not an extra independent axiom once Maxwell's equations are adopted in their full form. It follows directly.

Take the divergence of (4):

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t}. \quad (56)$$

The left-hand side vanishes identically:

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\varepsilon_0} \right). \quad (57)$$

Hence

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.} \quad (58)$$

This is the **continuity equation**.

Remark 9.1. *In integral form, the continuity equation is*

$$\frac{d}{dt} \iiint_V \rho dV = -\oint_V \mathbf{J} \cdot d\mathbf{a}. \quad (59)$$

So the rate of decrease of total charge in a volume equals the outward current through its boundary.

10 The Homogeneous Equations and Potentials

The equations

$$\nabla \cdot \mathbf{B} = 0, \quad (60)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (61)$$

are called the **homogeneous Maxwell equations**.

10.1 Vector Potential

Since $\nabla \cdot \mathbf{B} = 0$, one may write

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (62)$$

10.2 Scalar Potential

Substitute into Faraday's law:

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} \mathbf{B} = -\nabla \times \frac{\partial \mathbf{A}}{\partial t}. \quad (63)$$

Thus

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad (64)$$

Any curl-free vector field is locally a gradient, so

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla \phi, \quad (65)$$

or

$$\boxed{\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}.} \quad (66)$$

10.3 Gauge Freedom

The potentials are not unique. For any smooth scalar function $\chi(\mathbf{r}, t)$,

$$\mathbf{A}' = \mathbf{A} + \nabla \chi, \quad (67)$$

$$\phi' = \phi - \frac{\partial \chi}{\partial t}, \quad (68)$$

produce the same fields.

10.4 Lorenz Gauge

Choose

$$\nabla \cdot \mathbf{A} + \mu_0 \varepsilon_0 \frac{\partial \phi}{\partial t} = 0. \quad (69)$$

Then Maxwell's equations reduce to wave equations for the potentials:

$$\nabla^2 \phi - \mu_0 \varepsilon_0 \frac{\partial^2 \phi}{\partial t^2} = -\frac{\rho}{\varepsilon_0}, \quad (70)$$

$$\nabla^2 \mathbf{A} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}. \quad (71)$$

10.5 Coulomb Gauge

Another common choice is

$$\nabla \cdot \mathbf{A} = 0. \quad (72)$$

In electrostatics this is particularly natural.

11 Electrostatics and Magnetostatics as Limits

11.1 Electrostatics

If $\frac{\partial \mathbf{B}}{\partial t} = 0$ and charges are static, then

$$\nabla \times \mathbf{E} = 0. \quad (73)$$

Hence

$$\mathbf{E} = -\nabla\phi. \quad (74)$$

Substituting into Gauss's law,

$$\nabla \cdot (-\nabla\phi) = \frac{\rho}{\varepsilon_0}, \quad (75)$$

so

$$\boxed{\nabla^2\phi = -\frac{\rho}{\varepsilon_0}.} \quad (76)$$

This is Poisson's equation.

In charge-free regions,

$$\nabla^2\phi = 0, \quad (77)$$

which is Laplace's equation.

11.2 Magnetostatics

If fields are time-independent, then

$$\nabla \cdot \mathbf{B} = 0, \quad (78)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (79)$$

Using $\mathbf{B} = \nabla \times \mathbf{A}$ and Coulomb gauge $\nabla \cdot \mathbf{A} = 0$,

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J}. \quad (80)$$

Then by (11),

$$-\nabla^2 \mathbf{A} = \mu_0 \mathbf{J}. \quad (81)$$

Hence

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}. \quad (82)$$

This leads to the magnetostatic potential solution

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (83)$$

Then

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (84)$$

leads to the Biot–Savart law.

12 Electromagnetic Waves in Vacuum

Set

$$\rho = 0, \quad \mathbf{J} = 0. \quad (85)$$

Then

$$\nabla \cdot \mathbf{E} = 0, \quad (86)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (87)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (88)$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (89)$$

12.1 Wave Equation for $\mathbf{E}$

Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t} \nabla \times \mathbf{B}. \quad (90)$$

Using Ampère-Maxwell in vacuum,

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (91)$$

Apply (11) and $\nabla \cdot \mathbf{E} = 0$:

$$-\nabla^2 \mathbf{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (92)$$

Therefore

$$\boxed{\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.} \quad (93)$$

12.2 Wave Equation for $\mathbf{B}$

Similarly,

$$\boxed{\nabla^2 \mathbf{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0.} \quad (94)$$

12.3 Wave Speed

Comparing with the standard wave equation,

$$\nabla^2 \psi - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0, \quad (95)$$

we identify

$$v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c. \quad (96)$$

This was Maxwell's profound prediction: *light is an electromagnetic wave.*

12.4 Plane Wave Solutions

Take

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}, \quad (97)$$

$$\mathbf{B}(\mathbf{r}, t) = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}. \quad (98)$$

Then

$$\mathbf{k} \cdot \mathbf{E}_0 = 0, \quad (99)$$

$$\mathbf{k} \cdot \mathbf{B}_0 = 0, \quad (100)$$

so the waves are transverse. Also,

$$\mathbf{B}_0 = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}_0, \quad (101)$$

and since $\omega = ck$,

$$B_0 = \frac{E_0}{c}. \quad (102)$$

Thus $\mathbf{E}$, $\mathbf{B}$, and the direction of propagation are mutually perpendicular.

13 Energy Conservation: Poynting's Theorem

Start with

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad (103)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (104)$$

Dot the first with $\mathbf{E}/\mu_0$:

$$\frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) = \mathbf{E} \cdot \mathbf{J} + \varepsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}. \quad (105)$$

Dot the second with $\mathbf{B}/\mu_0$:

$$\frac{1}{\mu_0} \mathbf{B} \cdot (\nabla \times \mathbf{E}) = -\frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t}. \quad (106)$$

Using identity (12),

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}). \quad (107)$$

After rearranging,

$$\frac{\partial}{\partial t} \left(\frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \right) + \nabla \cdot \left(\frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \right) = -\mathbf{J} \cdot \mathbf{E}. \quad (108)$$

Define

$$u = \frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2, \quad (109)$$

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (110)$$

Then

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}.} \quad (111)$$

This is the **Poynting theorem**. Here:

- u is the electromagnetic energy density,
- $\mathbf{S}$ is the **Poynting vector**, the energy flux density.

14 Momentum and the Maxwell Stress Tensor

The fields also carry momentum. The electromagnetic momentum density in vacuum is

$$\mathbf{g} = \varepsilon_0 \mathbf{E} \times \mathbf{B}. \quad (112)$$

The corresponding stress tensor is

$$T_{ij} = \varepsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right). \quad (113)$$

Then one can derive the momentum conservation law

$$\frac{\partial g_i}{\partial t} + \partial_j T_{ij} = -f_i, \quad (114)$$

where

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B} \quad (115)$$

is the Lorentz force density on matter.

This shows that the electromagnetic field carries not only energy but also momentum and stress.

15 Boundary Conditions

Maxwell's equations imply discontinuity relations across interfaces.

Let $\hat{\mathbf{n}}$ be the unit normal from medium 1 to medium 2.

15.1 Normal Component of $\mathbf{E}$

Apply Gauss's law to a thin pillbox crossing the boundary:

$$\hat{\mathbf{n}} \cdot (\mathbf{E}_2 - \mathbf{E}_1) = \frac{\sigma}{\varepsilon_0}, \quad (116)$$

where σ is the free surface charge density.

15.2 Normal Component of $\mathbf{B}$

Using Gauss's law for magnetism,

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0. \quad (117)$$

15.3 Tangential Component of $\mathbf{E}$

Apply Faraday's law to a tiny rectangular loop:

$$\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}. \quad (118)$$

15.4 Tangential Component of $\mathbf{B}$

Apply the Ampère-Maxwell law to a tiny loop. If $\mathbf{K}$ is the surface current density,

$$\hat{\mathbf{n}} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0 \mathbf{K}. \quad (119)$$

These are the standard vacuum boundary conditions.

16 Maxwell's Equations in Matter

In materials, it is convenient to separate free and bound charges. Define polarization $\mathbf{P}$ and magnetization $\mathbf{M}$, and then define

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \quad (120)$$

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (121)$$

Then the macroscopic Maxwell equations are

$$\nabla \cdot \mathbf{D} = \rho_f, \quad (122)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (123)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (124)$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}. \quad (125)$$

For linear, isotropic, homogeneous media,

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}. \quad (126)$$

Then the wave speed in the medium is

$$v = \frac{1}{\sqrt{\mu\varepsilon}}. \quad (127)$$

17 Covariant Formulation

Special relativity reveals the deepest structure of Maxwell's equations.

Define spacetime coordinates

$$x^\mu = (ct, x, y, z), \quad (128)$$

the four-current

$$J^\mu = (c\rho, \mathbf{J}), \quad (129)$$

and the electromagnetic field tensor

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}. \quad (130)$$

Then the inhomogeneous Maxwell equations become

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu. \quad (131)$$

The homogeneous equations become

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0. \quad (132)$$

Charge conservation follows immediately:

$$\partial_\nu J^\nu = 0. \quad (133)$$

This covariant form shows that electric and magnetic fields are different aspects of one relativistic field.

18 Worked Examples

18.1 Example 1: Flux of a Point Charge

Example 18.1. *Show directly that the electric flux through any sphere centered on a point charge q is q/ε_0 .*

Proof. For a sphere of radius R ,

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_0 R^2} \hat{\mathbf{r}}. \quad (134)$$

Since $d\mathbf{a} = \hat{\mathbf{r}} dA$,

$$\mathbf{E} \cdot d\mathbf{a} = \frac{q}{4\pi\varepsilon_0 R^2} dA. \quad (135)$$

Integrate:

$$\Phi_{E=S} \mathbf{E} \cdot d\mathbf{a} = \frac{q}{4\pi\varepsilon_0 R^2} \int_S dA = \frac{q}{4\pi\varepsilon_0 R^2} (4\pi R^2) = \frac{q}{\varepsilon_0}. \quad (136)$$

□

18.2 Example 2: Infinite Straight Wire

Example 18.2. *Find the magnetic field of an infinite straight wire carrying current I .*

By cylindrical symmetry, $\mathbf{B}$ circles the wire and has constant magnitude on a circle of radius r :

$$\mathbf{B} = B(r) \hat{\phi}. \quad (137)$$

Choose an Amperian loop of radius r . Then

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = B(r) \oint_C d\ell = B(r)(2\pi r). \quad (138)$$

Ampère's law gives

$$B(r)(2\pi r) = \mu_0 I. \quad (139)$$

Hence

$$\boxed{B(r) = \frac{\mu_0 I}{2\pi r}.} \quad (140)$$

18.3 Example 3: Charging Capacitor and Displacement Current

Example 18.3. *Show that the displacement current between capacitor plates equals the conduction current in the wire for an ideal parallel-plate capacitor.*

Let the plates have area A , separation small enough to neglect fringing, and charge $Q(t)$ on the positive plate. Then

$$E(t) = \frac{Q(t)}{\varepsilon_0 A}. \quad (141)$$

Electric flux through a plate-spanning surface is

$$\Phi_E = EA = \frac{Q(t)}{\varepsilon_0}. \quad (142)$$

Thus

$$I_D = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \frac{d}{dt} \left(\frac{Q}{\varepsilon_0} \right) = \frac{dQ}{dt} = I. \quad (143)$$

So

$$\boxed{I_D = I.} \quad (144)$$

This is the precise content of Maxwell's correction in the capacitor problem.

18.4 Example 4: Plane Electromagnetic Wave

Example 18.4. *Let*

$$\mathbf{E}(z, t) = E_0 \cos(kz - \omega t) \hat{\mathbf{i}}. \quad (145)$$

Find $\mathbf{B}$ for a wave traveling in the $+z$ direction.

Because propagation is along $\hat{\mathbf{k}}$ and $\mathbf{E}$ is along $\hat{\mathbf{i}}$, $\mathbf{B}$ must be along $\hat{\mathbf{j}}$. Also $B_0 = E_0/c$, so

$$\boxed{\mathbf{B}(z, t) = \frac{E_0}{c} \cos(kz - \omega t) \hat{\mathbf{j}}.} \quad (146)$$

Then

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} = \frac{E_0^2}{\mu_0 c} \cos^2(kz - \omega t) \hat{\mathbf{k}}. \quad (147)$$

18.5 Example 5: Boundary Condition at a Conducting Surface

Example 18.5. *For a perfect conductor in electrostatic equilibrium, show that the electric field just outside the surface is normal to the conductor and equals σ/ε_0 in magnitude.*

Inside a perfect conductor in electrostatics,

$$\mathbf{E}_{\text{inside}} = 0. \quad (148)$$

Tangential continuity from Faraday's law gives

$$\hat{\mathbf{n}} \times (\mathbf{E}_{\text{out}} - 0) = 0, \quad (149)$$

so $\mathbf{E}_{\text{out}}$ has no tangential component. Using the pillbox result,

$$\hat{\mathbf{n}} \cdot \mathbf{E}_{\text{out}} = \frac{\sigma}{\varepsilon_0}. \quad (150)$$

Hence

$$\boxed{\mathbf{E}_{\text{out}} = \frac{\sigma}{\varepsilon_0} \hat{\mathbf{n}}.} \quad (151)$$

19 Boundary-Value Problems: The Electrostatic View

Electrostatics often reduces to solving Poisson or Laplace equations with boundary data.

19.1 Poisson's Equation

$$\nabla^2 \phi = -\frac{\rho}{\varepsilon_0}. \quad (152)$$

19.2 Laplace's Equation

In charge-free regions,

$$\nabla^2 \phi = 0. \quad (153)$$

19.3 Uniqueness Theorem

Theorem 19.1 (Electrostatic uniqueness). *If the potential ϕ is specified on the boundary of a region, then the solution to Poisson's equation inside the region is unique.*

Idea of proof. Suppose ϕ_1 and ϕ_2 are two solutions with the same boundary values. Let

$$u = \phi_1 - \phi_2. \quad (154)$$

Then

$$\nabla^2 u = 0 \quad (155)$$

in the region, and $u = 0$ on the boundary.

Multiply by u and integrate:

$$\iiint_V u \nabla^2 u \, dV = 0. \quad (156)$$

Using integration by parts,

$$\iiint_V |\nabla u|^2 \, dV = \oint_{\partial V} u \frac{\partial u}{\partial n} \, da. \quad (157)$$

Since $u = 0$ on the boundary, the right-hand side vanishes. Thus

$$\iiint_V |\nabla u|^2 \, dV = 0. \quad (158)$$

So $\nabla u = 0$ everywhere, meaning u is constant. Since $u = 0$ on the boundary, $u = 0$ everywhere. Therefore $\phi_1 = \phi_2$. $\square$

This theorem is fundamental for methods such as separation of variables, Green's functions, and the method of images.

20 Conceptual Summary of the Four Equations

Meaning of each equation

1. Gauss's law for electricity:

Electric charges are sources of electric field.

2. Gauss's law for magnetism:

Magnetic field has no isolated sources in classical electromagnetism.

3. Faraday's law:

Time-varying magnetic fields induce circulating electric fields.

4. Ampère-Maxwell law:

Currents and time-varying electric fields induce circulating magnetic fields.

Taken together, the equations say that electric and magnetic fields are dynamically coupled and self-sustaining: a changing $\mathbf{E}$ creates $\mathbf{B}$, and a changing $\mathbf{B}$ creates $\mathbf{E}$. This mutual induction is the mechanism behind electromagnetic waves.

21 A Short Logical Flowchart

22 Conclusion

Maxwell's equations are among the most profound equations in physics because they unify:

- electrostatics,
- magnetostatics,
- induction,
- conservation of charge,
- light and electromagnetic waves,
- energy and momentum transport in fields,
- and, in covariant form, special relativity.

The integral forms give global statements about flux and circulation; the differential forms give local field laws. Their mathematical consequences are rich and far-reaching: they produce wave equations, potentials, boundary conditions, uniqueness results, and conservation laws.

At a deeper level, Maxwell's equations are the classical field equations of the electromagnetic interaction.

The static electric and magnetic theories of Coulomb, Ampère, and Faraday become, through Maxwell, one dynamical theory of fields.